

Multiple Choice (10 marks)

1. If your training loss increases with number of epochs, which of the following could be a possible issue with the learning process?
 - (a) Regularization is too low and model is overfitting
 - (b) Regularization is too high and model is underfitting
 - (c) Step size is too large
 - (d) Step size is too small
2. Statement 1| The BLEU metric uses precision, while the ROGUE metric uses recall.
Statement 2| Hidden markov models were frequently used to model English sentences.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. Statement 1| Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2| DenseNets usually cost more memory than ResNets.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$? [note: as the product of the conditional probabilities]
 - (a) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$
 - (b) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$
 - (c) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: The convergence of a neural network is indeed influenced by the learning rate used during training.
7. True or False: In a Bayesian network with nodes H, U, P, and W, a total of 15 independent parameters are needed without assumptions of independence.
8. True or False: High entropy suggests that the partitions in classification are pure, indicating a clear separation of classes.
9. True or False: The original ResNet employs self-attention mechanisms, similar to those used in Transformers.
10. True or False: $P(A, B, C)$ can be expressed as $P(A, B | C) * P(C)$ for Boolean random variables A, B, and C without independence assumptions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that matches a word at the beginning of a string.
12. Write a function to locate the left insertion point for a specified value in sorted order.

End of Paper